

Discretization

❑ Three types of attributes

- ❑ Nominal—values from an unordered set, e.g., color, profession
- ❑ Ordinal—values from an ordered set, e.g., military or academic rank
- ❑ Numeric—real numbers, e.g., integer or real numbers

❑ Discretization: Divide the range of a continuous attribute into intervals

- ❑ Interval labels can then be used to replace actual data values
- ❑ Reduce data size by discretization
- ❑ Supervised vs. unsupervised
- ❑ Split (top-down) vs. merge (bottom-up)
- ❑ Discretization can be performed recursively on an attribute
- ❑ Prepare for further analysis, e.g., classification

Data Discretization Methods

- ❑ **Binning**

- ❑ Top-down split, unsupervised

- ❑ **Histogram analysis**

- ❑ Top-down split, unsupervised

- ❑ **Clustering analysis**

- ❑ Unsupervised, top-down split or bottom-up merge

- ❑ **Decision-tree analysis**

- ❑ Supervised, top-down split

- ❑ **Correlation (e.g., χ^2) analysis**

- ❑ Unsupervised, bottom-up merge

- ❑ **Note: All methods can be applied recursively**